

Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's network in 2024

17% Global Traffic Growth and Starlink's 3.3x Surge

+17%

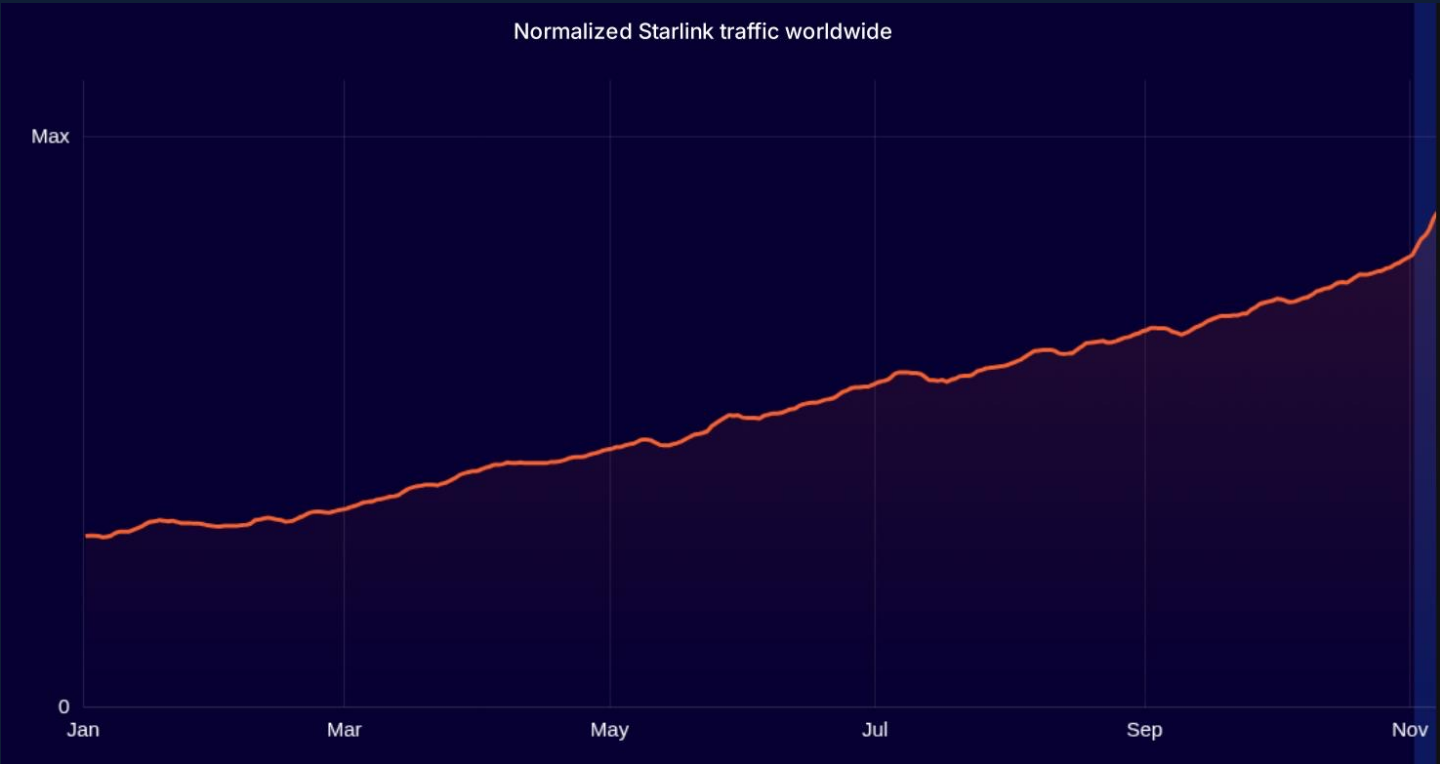
YoY Global Traffic Growth

3.3x

Starlink Traffic Surge

41%

Traffic from Mobile Devices



Key Insights

- Global Internet traffic grew 17% year-over-year in 2024, continuing the upward trend driven by increasing reliance on the Internet.
- SpaceX Starlink traffic surged 3.3x during 2024, expanding satellite connectivity to previously underserved locations.
- 41% of global traffic came from mobile devices, reflecting the continued importance of mobile connectivity worldwide.
- 33% of mobile traffic came from iOS devices; Android exceeded 95% share in some regions.
- Trend line starts mid-January to normalize for post-holiday activity.

Google Leads Top 10 Services as Generative AI Rises

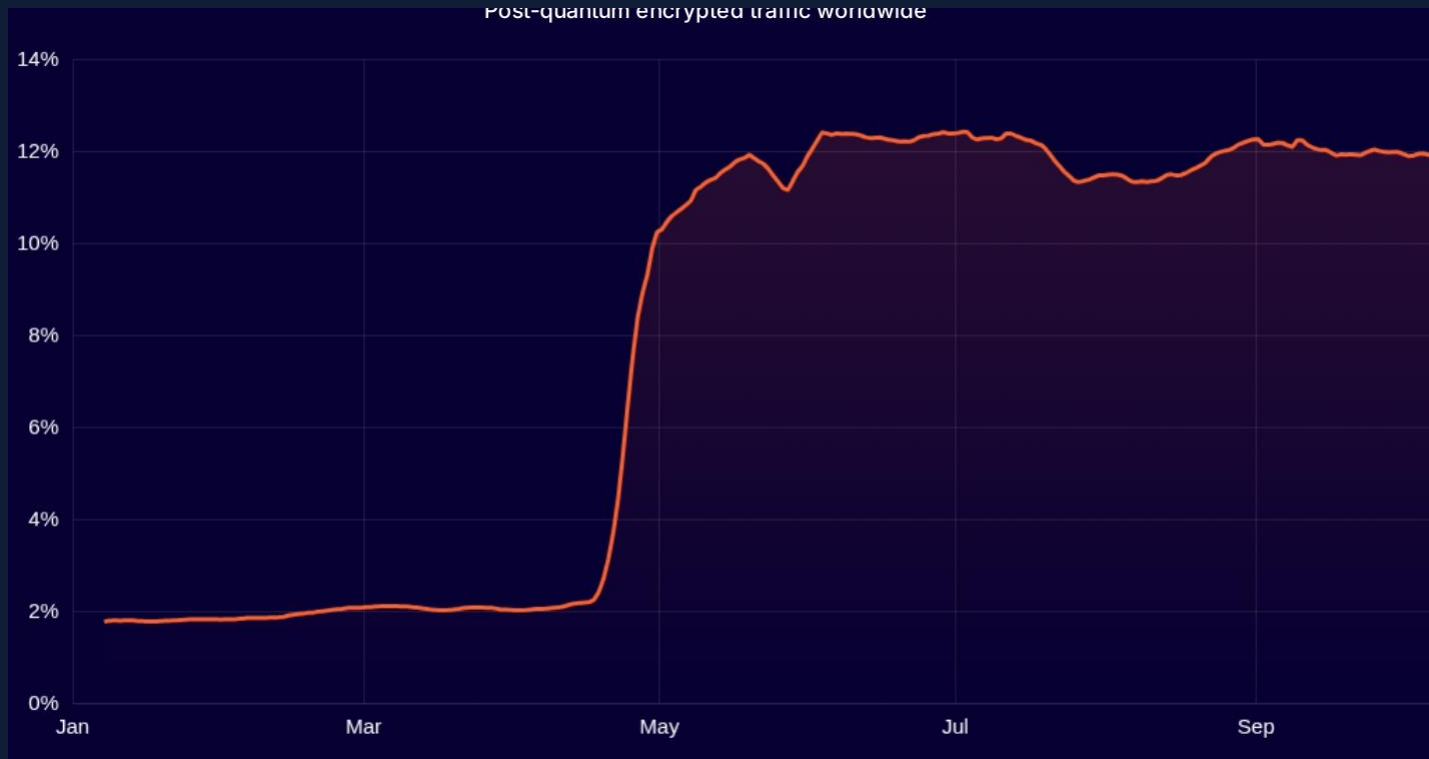
Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Generative AI: The Defining Trend of 2024

Generative AI continued its meteoric rise throughout 2024, while the Metaverse receded into the background. Rankings based on aggregate data from Cloudflare’s 1.1.1.1 DNS resolver.

Rank	Service
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Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3



2% → 13%

Post-Quantum TLS 1.3 Share

21%

HTTP/3 Adoption

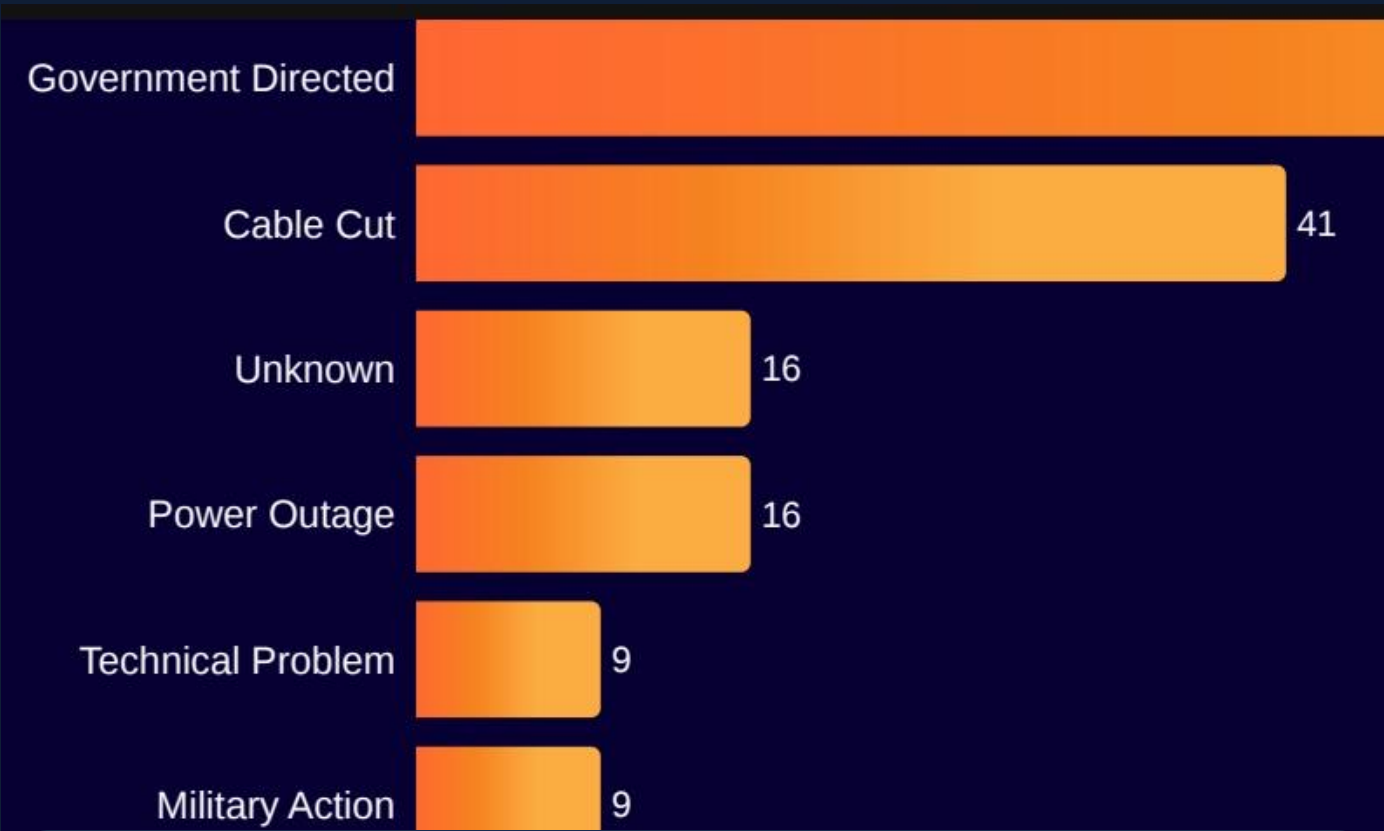
Go #1

API Client Language (~12%)

Technology Adoption Highlights

- Post-quantum encryption leapt from ~2% to 13% of TLS 1.3 traffic after Chrome 124 enabled it by default on April 17, 2024.
- Mozilla Firefox began rolling out post-quantum by default; Apple Safari started initial testing.
- HTTP/3 (QUIC-based) reached 21% adoption — providing faster connections, packet-loss mitigation, and encryption by default.
- Go surpassed other languages as the top API client language (~12%), followed by Node.js, Python (9.6%), Java (7.4%), and .NET (3.6%).
- Google dominates search engine referral traffic at 89%, followed by Yandex (3.1%), Baidu (2.7%), Bing (2.6%).
- Chrome remains the most popular browser globally.

225 Major Disruptions: Government Shutdowns Lead Outage Causes



Government-Directed Shutdowns Dominate

Government-directed shutdowns accounted for 116+ of 225 major disruptions (52%+), far exceeding all other causes combined. Cable cuts (41) were the second leading cause, followed by power outages and unknown factors (16 each). IPv6 adoption reached 28% of dual-stack traffic. Spain recorded the highest download speed among all countries measured.

Cause	Count
Government Directed	116+
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance	4
Cyber Attack	4
Natural Disaster	1
Fire	1

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

Traffic Mitigated

4.3%

Emails Were Malicious

68%

Bot Traffic from Top 10 Countries

43%

Malicious Emails with
Deceptive Links

Most Attacked Sector

Gambling & Games

The Gambling and Games sector was the most attacked industry category in 2024, reflecting the high-value nature of these platforms as targets.

Log4j Persistence

CVE-2021-44228

First disclosed December 2021, Log4j continues to be actively targeted more than two years later, underscoring the persistent nature of widely known vulnerabilities.

Email Threat Metrics

- 4.3% of emails processed were malicious
- 43% of malicious emails contained deceptive links
- .bar TLD originated the largest share of malicious and spam email

Routing Security Progress

RPKI (Resource Public Key Infrastructure) valid routes increased by 6.4%, an important step toward improved Internet routing security. 68% of global bot traffic originated from just the top 10 countries, highlighting geographic concentration of automated threats.

Five Takeaways Shaping the 2025 Internet Outlook

1

Traffic Growth Trajectory

Global Internet traffic grew 17% year-over-year, with Starlink surging 3.3x as satellite connectivity expanded to underserved regions. The growth trend shows no sign of slowing.

2

Post-Quantum Encryption Momentum

Post-quantum encryption leapt from ~2% to 13% of TLS 1.3 traffic following Chrome 124's default enablement. Expect Firefox and Safari adoption to accelerate this shift in 2025.

3

Connectivity Fragility from Government Actions

Government-directed shutdowns (116+) accounted for over half of 225 major Internet disruptions. Cable cuts (41) and power outages (16) remain persistent infrastructure risks.

4

Persistent Security Threats

6.5% of all traffic was mitigated; Gambling/Games was the most targeted sector. Log4j exploitation persists years after disclosure. 4.3% of emails were malicious, with 43% containing deceptive links.

5

Evolving Developer Ecosystem

Go became the top API client language (~12%), surpassing Node.js and Python (9.6%). HTTP/3 reached 21% adoption. The developer landscape is shifting toward newer, more efficient protocols and languages.

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